

Math 220B HW 2

Zih-Yu Hsieh

March 03, 2026

1 D

Problem 1

Let R be a semilocal ring and let M and N be two R -modules such that $M \otimes_R N = 0$. Prove or disprove that either $M = 0$ or $N = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, which is a semilocal ring because it only has finitely many maximal ideals (in fact, finitely many ideals since the ring is finite).

Now, consider the canonical projection $\pi_2 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\pi_3 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, notice that since $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$, one has the ideal $6\mathbb{Z} \subset \ker(\pi_2) = 2\mathbb{Z}$ and $6\mathbb{Z} \subset \ker(\pi_3) = 3\mathbb{Z}$. Hence, by Generalized First Isomorphism Theorem, with the canonical projection $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, the above two maps uniquely factor into surjective maps $\varphi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\varphi_3 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ as follows:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\pi_2} & \mathbb{Z}/2\mathbb{Z} \\ & \searrow \pi_6 & \nearrow \varphi_2 \\ & \mathbb{Z}/6\mathbb{Z} & \end{array} \qquad \begin{array}{ccc} \mathbb{Z} & \xrightarrow{\pi_3} & \mathbb{Z}/3\mathbb{Z} \\ & \searrow \pi_6 & \nearrow \varphi_3 \\ & \mathbb{Z}/6\mathbb{Z} & \end{array}$$

Hence, $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$ can be realized as R -modules, via the maps φ_2 and φ_3 (in particular, one has $\varphi_2(\bar{1}_6) = \bar{1}_2$ and $\varphi_3(\bar{1}_6) = \bar{1}_3$ as the maps based on the above commutative diagrams).

Now, consider the tensor $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z}$, we claim that it is 0: Recall that $2, 3 \in \mathbb{Z}$ are coprime, with Bezout's Lemma there exist integers $s, t \in \mathbb{Z}$, such that $1 = 2s + 3t$. Then, under the projection map $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, one knows there exist $\bar{s}_6, \bar{t}_6 \in \mathbb{Z}/6\mathbb{Z}$, such that the following holds:

$$\bar{1}_6 = \bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6 \tag{1.1}$$

Then, for any $\bar{a}_2 \in \mathbb{Z}/2\mathbb{Z}$ and $\bar{b}_3 \in \mathbb{Z}/3\mathbb{Z}$, one has the following:

$$\begin{aligned} \bar{a}_2 \otimes \bar{b}_3 &= \bar{1}_6 \cdot (\bar{a}_2 \otimes \bar{b}_3) = (\bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6) \cdot (\bar{a}_2 \otimes \bar{b}_3) \\ &= \bar{s}_6 \cdot ((\bar{2}_6 \cdot \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{3}_6 \cdot \bar{b}_3)) \\ &= \bar{s}_6 \cdot ((\bar{0}_2 \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{0}_3 \bar{b}_3)) \\ &= \bar{s}_6 \cdot (\bar{0}_2 \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes \bar{0}_3) \\ &= 0 \end{aligned} \tag{1.2}$$

(Note: By definition, one has $\bar{2}_6 \cdot \bar{a}_2 := \varphi_2(\bar{2}_6)\bar{a}_2 = \bar{0}_2\bar{a}_2 = \bar{0}_2$, and same idea for $\mathbb{Z}/3\mathbb{Z}$).

Hence, this proves that $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z} = 0$. Yet, each $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z} \neq 0$ as R -module, which is a desired counterexample for the problem.

2 D

Problem 2

Let $0 \rightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \rightarrow 0$ be an exact sequence of R -modules. Assume that M' and M'' are finitely generated R -modules. Prove that M is a finitely generated R -modules. Explain if the converse is true.

Solution: We'll first prove the given statement, then prove the converse is false in general, by providing a counterexample.

I. Proof of Statement:

Given the given sequence is exact, then it implies $M' \xrightarrow{u} M$ is injective, hence $M' \cong \text{im}(u) \subset M$, WLOG we'll denote $\text{im}(u)$ as M' , and $M' \xrightarrow{u} M$ as an inclusion map, and recognize all elements in M' as in M also.

Since M', M'' are finitely-generated, there exists $x_1, \dots, x_n \in M'$ and $y_1, \dots, y_m \in M''$, such that $M' = \sum_{i=1}^n Rx_i$ and $M'' = \sum_{j=1}^m Ry_j$. Then, choose $z_1, \dots, z_m \in M$, such that for each index $j \in \{1, \dots, m\}$, one has $v(z_j) = y_j$. We claim that the collection $\{x_1, \dots, x_n, z_1, \dots, z_m\}$ generates M .

For any $x \in M$, since $v(x) \in M''$, there exists $b_1, \dots, b_m \in R$, such that the following holds:

$$v(x) = \sum_{j=1}^m b_j y_j = \sum_{j=1}^m b_j v(z_j) = v\left(\sum_{j=1}^m b_j z_j\right) \quad (2.1)$$

As a result, one has $v\left(x - \sum_{j=1}^m b_j z_j\right) = 0$, or $x - \sum_{j=1}^m b_j z_j \in \ker(v)$.

Now, using exactness of the sequence, one gets $\ker(v) = \text{im}(u) = M'$, hence there exists $a_1, \dots, a_n \in R$, such that $x - \sum_{j=1}^m b_j z_j = \sum_{i=1}^n a_i x_i$ by the finitely generated property of M' .

As a result, we have $x = \sum_{i=1}^n a_i x_i + \sum_{j=1}^m b_j z_j$, hence all element $x \in M$ can be expressed with some R -linear combination of $\{x_1, \dots, x_n, z_1, \dots, z_m\}$, showing M is finitely generated.

II. Counterexample of the Converse:

Let k be a field, consider the ring $R = k[x_1, x_2, \dots]$, which R is a polynomial ring over k , with (countably) infinite indeterminates. First, let's note that R consists of polynomials that are finite sums of monomials, and each monomial can involve only finite indeterminates.

Consider the ideal $I := (x_1, x_2, \dots)$ (ideal generated by all indeterminates). Then, $I, R, R/I$ are all R -modules, in particular R is a finitely generated R -module (since every $r \in R$ can be written as $r \cdot 1$, so R is generated by 1 as an R -module). Moreover, we naturally have the following exact sequence of R -linear maps:

$$0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$$

And it's simply because the image of the inclusion $I \hookrightarrow R$ is I , and the kernel of projection $R \twoheadrightarrow R/I$ is also I .

Now, we have a short exact sequence, such that the middle R -module is finitely generated. Yet, we claim that I is not a finitely generated R -module:

Suppose the contrary that I is a finitely generated R -module, then there exists $f_1, \dots, f_n \in I$, such that $I = \sum_{i=1}^n Rf_i = (f_1, \dots, f_n)$ as ideal. Based on our definition, there are only finitely many indeterminates involved in $f_1, \dots, f_n$, say $S = \{x_{i_1}, \dots, x_{i_k}\}$ (and say it's ordered such that $i_1 < \dots < i_k$).

Then, consider the element $x_{i_{k+1}} \in I$: By our assumption, there exists polynomials $g_1, \dots, g_n \in R$, such that $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$. Which, let $T = \{x_{j_1}, \dots, x_{j_l}\}$ be the (finite) indeterminates involved in the polynomials $g_1, \dots, g_n$. Which, if consider the set of indeterminates $S \cup T \cup \{x_{i_{k+1}}\} = \{x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}\}$ after reordering, one can realize the elements $x_{i_{k+1}}, f_1, \dots, f_n, g_1, \dots, g_n$ as elements in $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \hookrightarrow k[x_1, x_2, \dots]$, and inside this smaller subring, the equality $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$ still holds.

Now, consider the evaluation map $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \rightarrow k$, by $x_{i_{k+1}} \mapsto 1$, and all $x_{m_r} \mapsto 0$: Since we've chosen $x_{i_{k+1}}$, such that it's distinct from all x_{i_j} ($1 \leq j \leq k$), then one has x_{i_j} being some of the x_{m_r} , hence $x_{i_j} \mapsto 0$. Also, notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + \dots$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i , which is no longer with constant term). Hence, with each $x_{i_j} \mapsto 0$, $f_i \mapsto 0$ also (since it's a polynomial involving only $x_{i_1}, \dots, x_{i_k}$, and with no constant term).

As a result, one has the following:

$$x_{i_{k+1}} = \sum_{i=1}^n g_i f_i \mapsto 0 \quad (2.2)$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{i_{k+1}} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R -module.

Hence, we found an exact sequence of R -modules $0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$, such that the middle is finitely generated R -module, while the two sides are not guaranteed to be finitely generated (in particular, the left side is not guaranteed). This proves that the converse is in general false.

3 D

Problem 3

Let M and N be flat R -modules, where R is a commutative ring. Prove or disprove that $M \otimes_R N$ is flat R -module.

Solution: We'll prove that $M \otimes_R N$ is also a flat R -module.

Since the tensor functor $(_) \otimes_R (M \otimes_R N) : \mathbf{R}\text{-Mod} \rightarrow \mathbf{R}\text{-Mod}$ is a right exact functor, to show that $M \otimes_R N$ is flat (or the above tensor functor is exact), one only needs to prove it preserves injective R -linear maps.

Recall that for any R -module K, L, P , there is a unique R -linear isomorphism $\varphi_{KLP} : (K \otimes_R L) \otimes_R P \xrightarrow{\sim} K \otimes_R (L \otimes_R P)$ that satisfies $\varphi_{KLP}((k \otimes l) \otimes p) = k \otimes (l \otimes p)$ for all $k \in K, l \in L$, and $p \in P$.

Now, given an injective R -linear map $f : K \hookrightarrow K'$, since M is flat, then the exactness of the functor $(_) \otimes_R M$ guarantees the map $f \otimes \text{id}_M : K \otimes_R M \rightarrow K' \otimes_R M$ is injective (which the map is given by $(f \otimes \text{id}_M)(k \otimes m) = f(k) \otimes m$ for all $k \in K, m \in M$).

Similarly, since N is flat, then the exactness of the functor $(_) \otimes_R N$ guarantees the map $(f \otimes \text{id}_M) \otimes \text{id}_N : (K \otimes_R M) \otimes_R N \rightarrow (K' \otimes_R M) \otimes_R N$ is injective (which the map is given by $((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) = (f(k) \otimes m) \otimes n$ for all $k \in K$, $m \in M$, and $n \in N$).

Finally, consider the following diagram:

$$\begin{array}{ccc}
 K \otimes_R (M \otimes_R N) & \xrightarrow{f \otimes \text{id}_{M \otimes_R N}} & K' \otimes_R (M \otimes_R N) \\
 \varphi_{KMN}^{-1} \downarrow & & \uparrow \varphi_{K'MN} \\
 (K \otimes_R M) \otimes_R N & \xrightarrow{(f \otimes \text{id}_M) \otimes \text{id}_N} & (K' \otimes_R M) \otimes_R N
 \end{array}$$

Notice that the diagram commutes, because for each $k \in K$, $m \in M$, and $n \in N$, one has the following:

$$(f \otimes \text{id}_{M \otimes_R N})(k \otimes (m \otimes n)) = f(k) \otimes (m \otimes n) \quad (3.1)$$

$$\begin{aligned}
 & \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N) \circ \varphi_{KMN}^{-1}(k \otimes (m \otimes n)) \\
 &= \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) \\
 &= \varphi_{K'MN}((f(k) \otimes m) \otimes n) \\
 &= f(k) \otimes (m \otimes n)
 \end{aligned} \quad (3.2)$$

Then, since $K \otimes_R (M \otimes_R N)$ is generated by all elements of the form $k \otimes (m \otimes n)$, then the above two calculation shows that the two maps are the same, hence the diagram commutes.

As a result, since $\varphi_{K'MN}$, $(f \otimes \text{id}_M) \otimes \text{id}_N$, and φ_{KMN}^{-1} are all injective, then their composition – which results in $f \otimes \text{id}_{M \otimes_R N}$ – is injective. Therefore, the functor $(_) \otimes_R (M \otimes_R N)$ preserves injective maps, hence is exact. This shows that $M \otimes_R N$ is a flat module.

4 ND

Problem 4

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be a flat R -module. Prove or disprove that $M \otimes_R S$ is a flat S -module.

Solution: We'll prove that given M is a flat R -module, then $M \otimes_R S$ is a flat S -module.

For this, let's prove an exercise in Atiyah Macdonald Chapter 2:

Exercise: 2.15

Let A, B be commutative rings, let M be an A -module, P a B -module and N an (A, B) -bimodule (that is, N is simultaneously an A -module and B -module and the two structures are compatible in the sense that $a(xb) = (ax)b$ for all $a \in A$, $b \in B$, $x \in N$).

Then, $M \otimes_A N$ is naturally a B -module, $N \otimes_B P$ an A -module, and we have:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P) \quad (4.1)$$

Proof: We'll prove the three claims in sequence:

I. B -Module Structure on $M \otimes_A N$:

For any $b \in B$, $m \in M$ and $n \in N$, define the action $B \times (M \otimes_A N) \rightarrow M \otimes_A N$ as $b \cdot (m \otimes_A n) := m \otimes_A (nb)$ on each generator $m \otimes_A n$ (where nb is given by the B -module structure of N), we also define its action on any sum $\sum_{i=1}^k m_i \otimes_A n_i$ by sum of its action on individual $m_i \otimes_A n_i$. As a set map this is well-defined, let's check it does satisfy the B -module properties (which suffices to check for the generators of $M \otimes_A N$). Given any $b, b' \in B$, and any element $m \otimes_A n, m' \otimes_A n' \in M \otimes_A N$, we have the following:

$$\begin{aligned} 1. \quad (b + b') \cdot (m \otimes_A n) &= m \otimes_A (n(b + b')) = m \otimes_A (nb + nb') \\ &= m \otimes_A (nb) + m \otimes_A (nb') \end{aligned} \quad (4.2)$$

$$\begin{aligned} 2. \quad b \cdot (m \otimes_A n + m' \otimes_A n') &= b \cdot (m \otimes_A n) + b \cdot (m' \otimes_A n') \end{aligned} \quad (4.3)$$

(Note: This is by our definition, such that the action on the sum, is the sum of each individual generator after the action).

$$\begin{aligned} 3. \quad (bb') \cdot (m \otimes_A n) &= m \otimes_A (n(bb')) = m \otimes_A (n(b'b)) \\ &= m \otimes_A ((nb')b) = b \cdot (m \otimes_A (nb')) \\ &= b \cdot (b' \cdot (m \otimes_A n)) \end{aligned} \quad (4.4)$$

$$4. \quad 1_B \cdot (m \otimes_A n) = m \otimes_A (n1_B) = m \otimes_A n \quad (4.5)$$

These are all the requirements for being a B -module, hence $M \otimes_A N$ has a B -module structure (by its action on N).

II. A -Module Structure on $N \otimes_B P$:

Similarly, define $A \times (N \otimes_B P) \rightarrow N \otimes_B P$ by $a \cdot (n \otimes_B p) := (an) \otimes_B p$. Based on similar reasonings in I, it satisfies all the conditions for A -module, hence $N \otimes_B P$ now can be realized as A -module.

III. Isomorphism:

We aim to show that $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are isomorphic as both A and B -module. For this, we'll construct a B -linear map $(M \otimes_A N) \otimes_B P \rightarrow \square$

Since S is a bimodule (where it's both an R -module and S -module, while any $r \in R$, $s, t \in S$ has $(r \cdot t)s = (\varphi(r)t)s = \varphi(r)(ts) = r \cdot (ts)$, where the dot $\cdot$ represents the R -action on S), then one has the following isomorphism, for any S -module P :

$$(M \otimes_R S) \otimes_S P \cong M \otimes_R (S \otimes_S P) \quad (4.6)$$

Then, given any injective S -linear map $P \hookrightarrow P'$ between two S -modules, one has the following diagram:

5 ND

Problem 5

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that M is projective as an R -module. Prove or disprove that $M \otimes_R S$ is projective as S -module.

Solution: We'll prove that if M is projective as R -module, then $M \otimes_R S$ is projective as S -module.

Since M is projective as an R -module, there exists another R -module N , such that $M \oplus N = R^{\oplus I}$. Not sure if this is right, but if $M \oplus N = R^{\oplus I}$, then if $R^I \otimes_R S \cong (R \otimes_R S)^{\oplus I} \cong S^{\oplus I}$ as R -module. Then, the question becomes: are the two structures compatible? Namely the

6 ND

Problem 6

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be an injective R -module such that $M \otimes_R S \neq 0$. Prove or disprove that $M \otimes_R S$ is injective S -module.

Solution:

7 ND

Problem 7

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that S is flat R -algebra. Prove or disprove that φ is injective.

Solution:

8 ND

Problem 8

Let $\varphi : R \rightarrow S$ be an injective ring homomorphism such that S is a field. Prove or disprove that S is a flat R -algebra.

Solution:

9 ND

Problem 9

Let R be a dedekind domain and let $I \subset R$ be an ideal. Prove or disprove that I is a flat R -module.

Solution:

10 ND

Problem 10

Let R be a PID and let M be a finitely generated flat R -module. Prove or disprove that M is a free R -module.

Solution: Q: Over a PID, is there any flat module that's not projective?

11 ND

Problem 11

Let R be a PID and let M be an R -module. Prove that R is a submodule of an injective R -module.

Solution: The injective resolution problem...

12 ND

Problem 12

Let R be a commutative ring and let m be a given maximal ideal of R . Suppose M is an R -module such that $M_m = 0$. Prove or disprove that $M = 0$.

Solution:

13 ND

Problem 13

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be two ring homomorphisms such that S is R -flat and T is S -flat. Prove or disprove that T is R -flat.

Solution:

14 ND

Problem 14

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that φ can not be surjective.

Solution:

15 ND

Problem 15

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that ψ can not be surjective.

Solution:

16 ND

Problem 16

Let $R \xrightarrow{\varphi} S$ be ring homomorphism which is flat. Let M be an R -module such that $M \otimes_R S$ is torsion-free S -module. Is M torsion-free R -module?

Solution: